KASHISH SAINI

AI/ML Developer | Generative AI | Python

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PROFESSIONAL SUMMARY

B.Tech CSE graduate (CGPA 8.10/10) and AI/ML Intern building and benchmarking deep generative models (VAE, GAN, Diffusion) in PyTorch for synthetic data generation. Evaluated four VAE variants, DCGAN and DDPM on MNIST, CIFAR10, retail and chest X-ray images using FID, LPIPS and Inception Score, and measured downstream gains with YOLOv8 classifiers. Also built two full-stack LLM applications with FastAPI, React and Groq (Llama 3.3). Seeking an entry-level AI/ML Engineer or Generative AI Developer role.

TECHNICAL SKILLS

Generative AI / DL: PyTorch, VAE, Beta-VAE, CVAE, VQ-VAE, GAN (DCGAN), Diffusion (DDPM, DDIM), U-Net, Optuna
Evaluation / CV: FID, LPIPS, Inception Score, YOLOv8-cls, pseudo-labelling, data-scarcity experiments
LLM / Agents: Groq Cloud API (Llama 3.3 70B), LiteLLM, Strands Agents, Prompt Engineering
Languages: Python, Java (Core), JavaScript (ES6+), C, C++
Web / Backend: FastAPI, Django, REST APIs, React.js, Redux Toolkit, HTML5, CSS3
Databases / Tools: MySQL, SQLite, MongoDB, SQLAlchemy | Git, GitHub, Kaggle (T4 GPU), fp16 mixed precision
Core CS: Data Structures and Algorithms, Object-Oriented Programming

EXPERIENCE

AI/ML Intern, Synthetic Data Generation | Truminds Software Systems Pvt. Ltd., Gurgaon (On-site) | *Aug 2026 – Present*

- Built and benchmarked **4 VAE variants** (Vanilla, CVAE, Beta-VAE, VQ-VAE) in PyTorch on retail and chest X-ray images; tuned learning rate, latent size, beta and codebook parameters with **Optuna**, raising downstream YOLOv8 accuracy from 16.2% to 17.0%.
- Trained **DCGAN** and **DDPM (U-Net, EMA, DDIM sampling)** on MNIST and CIFAR10; stabilised GAN training (label smoothing, gradient clipping) after fixing a discriminator/generator collapse.
- Showed diffusion gave better image quality: mean **FID 88.2 (DDPM) vs 137.6 (DCGAN)**; on MNIST FID 13.2 vs 54-69; scarce-data DDPM reached CIFAR10 FID 110.2 using only 60 real images/class.
- Quantified data scarcity: training on 60 images/class raised VAE FID from **211.3 to 320.4 (+51.6%)**.
- Designed a synthetic-augmentation pipeline: pseudo-labelled generated images with YOLOv8n-cls (**99.0% MNIST, 92.2% CIFAR10**), compared LPIPS-based vs random selection (60 vs 180 images/class); LPIPS selection cut perceptual distance by **36-67%**, and 180 random images gave the best accuracy (**84.3-84.5%**).
- Optimised runs for a Kaggle T4 GPU budget with fp16 mixed precision, checkpoint scoring and seeded, reproducible comparisons documented in cross-notebook reports.

PROJECTS

StudyMate AI: Smart Student Assistant | Python, FastAPI, Strands Agents, Groq API (Llama 3.3 70B), LiteLLM, PyPDF2 | GitHub | Live Demo

- Built an AI study assistant with native Hinglish chat, multi-format document parsing (PDF, DOCX, images) and live camera-capture context.
- Engineered an asynchronous pipeline with LiteLLM and Strands Agents, resolving token-layout and multi-format extraction bottlenecks.

HCP Interaction AI CRM: Pharma Automation Tool | Python, FastAPI, React.js, Redux Toolkit, Groq Cloud API, SQLAlchemy | GitHub | Live Demo

- Built a pharma CRM tool that parses raw HCP meeting logs into structured medical fields with automated sentiment analysis.
- Used Redux Toolkit state synchronisation for high-frequency JSON streams, removing UI re-rendering lag during clinical log updates.

EDUCATION

B.Tech in Computer Science and Engineering | RV Institute of Technology, Bijnor (AKTU) | 2022 – 2026 | **CGPA: 8.10 / 10**

TRAINING

React.js Trainee (Aug–Sept 2025), Web Designing Trainee (Sept–Oct 2024), Java Core Trainee (Sept 2023) | XTRUDE Engineers Pvt. Ltd.